

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/230913533>

QSO+Galaxy Association and Discrepant Redshifts in NEQ3

Article in *The Astrophysical Journal* · December 2008

DOI: 10.1086/383346 · Source: arXiv

CITATIONS

3

READS

86

2 authors:



[Cristian Migueo Gutiérrez](#)

Universidad de América

120 PUBLICATIONS 2,701 CITATIONS

SEE PROFILE



[Martín López Corredoira](#)

Instituto de Astrofísica de Canarias

183 PUBLICATIONS 7,201 CITATIONS

SEE PROFILE

QSO+Galaxy association and discrepant redshifts in NEQ3

C. M. Gutiérrez

Instituto de Astrofísica de Canarias, E-38205, La Laguna, Tenerife, Spain

`cgc@ll.iac.es`

and

M. López-Corredoira

Astronomisches Institut der Universität Basel. Venusstrasse 7. CH-4102 Binningen, Switzerland

ABSTRACT

Spectroscopy and deep imaging of the group NEQ3 are presented. This system is formed by three compact objects with relative separations of ~ 2.6 and ~ 2.8 arcsec, and a lenticular galaxy at ~ 17 arcsec from the geometric centre of the group. A diffuse filament is located on a line joining the three compact objects and the main galaxy. Analysis of these observations confirms the redshift previously known for three of the objects ($z = 0.1239$ for the main galaxy, and $z = 0.1935$ and 0.1939 for two of the compact objects). We have also determined the previously unknown redshift of the third compact object as $z = 0.2229$. Using the relative strength and width of the main spectral lines we have classified the compact objects as two HII galaxies and one QSO (the object at $z = 0.1935$). With cross-correlation techniques, we have tentatively estimated the redshift of the filament as $z = 0.19$ (although a weaker component also appears at $z = 0.12$) so that it is probably associated with the halo of the two compact objects at this redshift. The two objects at redshift ~ 0.19 represent possibly one of the more clear examples of starburst (and perhaps QSO activity) driven by interaction. These, and the relation between these two objects and the other two at (~ 0.12 and ~ 0.22) make the nature of this system intriguing, being difficult to explain the whole association on conventional scenarios.

Subject headings: galaxies: photometry, individual (NEQ3), cosmology: anomalous redshifts

1. Introduction

In the last three years we have started a programme to study several apparently interacting systems with different redshifts. These constitute the so-called problem of anomalous redshifts (e.g. Burbidge 1996 and references therein). The first system that we studied in detail is formed by NGC 7603, its companion galaxy NGC 7603B, the filament apparently connecting both galaxies, and two objects within it (López-Corredoira & Gutiérrez 2002). In that article we showed that the two objects in the filament have redshifts of 0.24 and 0.39 respectively. In López-Corredoira & Gutiérrez (2003) we have presented subsequent observations on these objects and detected others at high redshift in the same field. Simple probability computations indicate the unlikelihood that this configuration is merely a projection effect.

We continued our study of several other systems in different observing campaigns conducted during 2002. Here, we present our observations and analysis of the system NEQ3 (the designation was proposed by Sulentic & Arp 1976 and means Non Equilibrium System Number 3). Although a rather intriguing system, the only other study of it, surprisingly, has been conducted by Arp (1977). The system is centred at $\text{RA}(\text{J2000}) = 12^{\text{h}} 09^{\text{m}} 51.6^{\text{s}}$, $\text{Dec.}(\text{J2000}) = +39^{\circ} 31' 29''$ at 10.8 arcmin from the exhaustively studied large active galaxy NGC 4151. Other interesting associations were found by Arp in the region around this galaxy. The system comprises three compact objects with relative separations of 2.6 and 2.8 arcsec with respect to the central object, all of them lying along the minor axis of an apparent lenticular galaxy at ~ 17 arcsec. Arp measured a redshift of 0.12 for the main galaxy and 0.19 for two of the compact objects. Arp described the objects as compact with emission lines. A filament is situated along the line connecting the main galaxy and the three compact objects.

All the above facts attracted our attention and made this system one of our top priorities. The goals of our study were to ascertain: i) the redshift of the third compact object, ii) the nature of these objects, iii) the redshift of the mentioned filament, iv) whether there are other objects with anomalous redshifts in the near field, v) whether there is evidence for distortions in the profile of the lenticular galaxy, vi) how likely it is that this optical association is merely a projection effect, and vii) what physical mechanisms could originate a system like this.

In this article we present the first results of this study. A more detailed analysis covering the photometry and velocity radial profile of the main galaxy and spectra of other objects in the field is in progress and will be presented in a future paper.

2. Observations

The observations presented here comprise broad band imaging and long-slit spectroscopy with intermediate resolution. These observations were taken at the NOT¹ (on 2002 December 2) and WHT² (on 2002 December 28–29). We follow a standard reduction method explained in López-Corredoira & Gutiérrez (2003). Table 1 presents a summary of the observations. We observed the field in the Sloan r' filter. We used the NOT with the ALFOSC instrument, which has a field of view ~ 6 arcmin with a spatial sampling of 0.188 arcsecs. The r' band was relatively calibrated using an image taken with the IAC 80 telescope³ and corrected to the SDSS r' filter through the relations given in Smith et al. (2002).

The spectra were taken with the red arm of the ISIS instrument and the R158R grism at the WHT. This configuration gives a sampling of $1.63 \text{ \AA}$ and a resolution, measured from the width of the spectral arc lines, of $\sim 8 \text{ \AA}$ for a slit width of 1.2 arcsec. The spectra have a useful range between ~ 3900 and $\sim 9600 \text{ \AA}$.

Figure 1 presents a false-colour scale of our image in the r' band. Superimposed are the positions in which we positioned the slits. These positions were chosen to cross the three compact objects, the main galaxy, the filaments within them and several other promising objects in the field. For clarity, the objects analysed here have been arbitrary labelled from 1 to 4.

3. Analysis

Figure 2 shows the system in detail. Objects 1, 2 and 3 appear rather compact, although two small distortions (see also Figure 1) in object 2, seem to point in the direction of the two close companions. Unfortunately our ability to detect small-scale structure is severely restricted by poor seeing; for example, the seeing (1.4 arcsec) in the r' band corresponds to $\sim 4 \text{ kpc}$ at a cosmological redshift of $z = 0.19$. We see how the system is surrounded by a diffuse emission. We have delineated this emission down to isophote $26.8 \text{ mag arcsec}^{-2}$. We also discern a filament (previously noted by Arp) along the line of the minor axis of object

¹The Nordic Optical Telescope (NOT) is jointly operated on the island of La Palma by Denmark, Finland, Iceland, Norway, and Sweden, in Spain’s Roque de los Muchachos Observatory of the Instituto de Astrofísica de Canarias (IAC).

²The William Herschel Telescope (WHT) is operated by the Isaac Newton Group and the IAC in Spain’s Roque de los Muchachos Observatory.

³The IAC-80 telescope is located at Spain’s Teide Observatory (Tenerife) and is operated by the IAC.

4. It seems also that the filament decreases in brightness from object 1–2 to object 4, which looks rather symmetric with a morphological shape typical of a lenticular (S0/Sa) galaxy, its spectrum presenting only absorption lines characteristic of an old population (Ca H, Ca K, MgI ($\lambda 5180$), NaI($\lambda 5892$)). Using them we have estimated a redshift of 0.1239 ± 0.0005 for this galaxy. The only signs of distortion are the filament in the direction to objects 1–3, and an extended emission feature to the north that seems to enclose a weak object ($r' = 21.8$ mag) situated at ~ 13 arcsec. We have determined the magnitudes of the four objects using the SExtractor software (Bertin & Arnouts 1996). The magnitudes of objects 1–4 in the r' band are respectively 19.8, 19.6, 20.2 and 17.3. As a consequence of the proximity of the three objects and the existence of diffuse emission around them, the uncertainty in the quoted magnitudes should be ~ 0.1 – 0.2 mag.

The spectra of objects 1, 2 and 3 are dominated by emission lines whose identification allow us to determine their redshift unambiguously. The main features are the lines of the hydrogen Balmer series, and the lines of OII and OIII. Other minor features such as SI, NI, NII and HeI are also detected. Figure 3, shows the spectra (corrected of redshift) around the lines $H\beta$ and $H\alpha$. Table 2 presents the equivalent widths of each of these lines. The positions of all of these lines is consistent within the uncertainty in our spectral resolution. The resulting redshifts are 0.1935 ± 0.0002 , 0.1939 ± 0.0005 and 0.2229 ± 0.0002 for objects 1, 2 and 3 respectively. Tentatively we have identified some possible absorption features in the spectra of the three objects which could be associated to the host or normal underlying stellar population.

Object 1 has a typical broad band line spectrum, while objects 2 and 3 have only narrow emission lines. They have strong star formation ($EW(H\alpha + NII) = 100$ and 70 mÅ respectively; Carter et al. 2001). We have classified the emission line objects using the diagnostics considered by Verón, Gonçalves, & Verón-Cetty (1997) based on the line ratios of $OIII(\lambda 5007)/H\beta$ and $NII(\lambda 6584)/H\alpha$. Because of the spectral proximity of the lines involved in each ratio, the effects of reddening are minimized and the unaccuracies on the determination of the continuum largely cancel out. According to these criteria, object 1 is a QSO/Seyfert 1 galaxy, while objects 2 and 3 are HII galaxies.

The spectrum of the filament is very noisy and does not have any obvious feature. However, using the subroutine *fxcor* of IRAF⁴ with a field bright star as template, we tentatively identify a maximum in the cross-correlation function which corresponds approximately to a redshift of 0.19. A secondary maxima appears also at $z = 0.12$, but we do not exclude

⁴IRAF is the Image Reduction and Analysis Facility, written and supported by the IRAF programming group at the National Optical Astronomy Observatories (NOAO) in Tucson, Arizona.

possibly contamination from the main galaxy.

4. Discussion

As in other cases of anomalous redshift, the optical configuration of the system formed by NEQ3 (objects 1 to 3), the close lenticular galaxy (object 4) and the additional features such as the filament apparently connecting them, and the diffuse emission in the northern direction seem to be clear indications of proximity and interaction. As discussed in López-Corredoira & Gutiérrez (2003), examples like this, in which galaxies interact through filaments and show distortions in the halos, are relative common. An interpretation that explains the configuration as equivalent to other systems in interaction would be clearly preferred over one in which the configuration is purely a projection effect.

Objects 1 and 2 have redshifts of $z = 0.1935$ and 0.1939 respectively and are separated by 2.8 arcsec. Considering the redshift as indicative of cosmological distance, the above numbers are translated into a difference of $\sim 100 \pm 200 \text{ km s}^{-1}$ in velocity and a separation of $\sim 8 \text{ kpc}$. Bahcall et al. (1997), in a study with the *Hubble Space Telescope*, found that most of the detected QSO hosts have a galaxy at less than 25 kpc. This companion tends to be an elliptical galaxy for radio-loud QSOs, and an elliptical or spiral galaxy for radio-quiet QSOs. Canalizo & Stockton (2001) have studied the environment of nine objects with intermediate colors between ultra luminous infrared galaxies and QSOs and have found close companion in some of them and morphological evidence of interaction in eight of them. In general the companion show a mixture of young and old stellar population but are not star forming galaxies. In that sample one of the system which perhaps resemble more NEQ3 is IRAS00275-2859 which shows a secondary nucleus (or a giant HII region) at $\sim 2.7 \text{ arcsec}$ from the QSO and a tail with several clumps extending 12 arcsec. Differently from the cases studied by these authors, in NEQ3 the two objects (the QSO and the HII galaxy) at redshift ~ 0.19 have similar magnitudes in the r band.

The association of objects 1 and 2 is particularly interesting because it is one of the cases with a smaller angular separation, the galaxy is undergoing an intense burst of star formation, and there is a diffuse filament possibly associated with this pair. All of these facts seem to indicate a strong interaction between the QSO and the HII galaxy. This interaction could be the responsible of the intense star formation in object 2 and perhaps also of the QSO activity (Stockton 1982).

In a conventional scenario the role of objects 3 and 4 is unclear. For instance, could the difference in redshift between objects 1 and 2, and object 3 (~ 0.03) be produced by a

difference in peculiar velocity? The difference would be 9000 km s^{-1} . As far as we know, interactions between galaxies with such differences in velocity have not been observed and it would be difficult to explain in the framework of models of galaxy formation. In favour of the physical relation between the pair of objects 1–2 and object 3 is the main halo which seems surround the three objects uniformly and the slight elongation (see Figures 1 and 2) of object 2 through objects 1 and 3.

We have discarded the possibility of gravitational lensing produced by object 4, because the combination of amplification and separation between the sources and the lens would require much more massive lensing structures such as the cores of galaxy clusters. This amplification by galaxy clusters has been used by Ellis et al. (2001) to detect high redshift galaxies.

Finally, is it just a merely projection effect the proximity of object 4? Is this proximity related with the presence of the possible filament extending outwards in the direction of the three compact objects? From the galaxy counts by Metcalfe et al. (1991) and assuming Poission statistics, we have made a simple probability calculation. Based on that, Table 3 presents an estimation of the integrated number of galaxies with magnitudes up the one of objects 2, 3 and 4. We start the computation with the pair of objects at $z \sim 0.19$, and compute the probability (P_1) to find an object as bright as 3 (which density is N_3 in a circle of ~ 3 arcsec of radius (area A_1). Assuming Poisson statistics (and onsidering that the density of each kind of objects in the area considered is very low) we have $P_1 = N_3 \times A_1 \sim 4 \times 10^{-3}$. The probability to have object 4 at the end of the filament (assuming roughly an area for this filament $A_2 \sim 15 \times 2.5 = 37.5 \text{ arcsec}^{-2}$) is $P_2 = N_4 \times A_2 \sim 5 \times 10^{-4}$. Then the total probability is: $P = P_1 * P_2 / 2 \sim 10^{-6}$. This low probability makes unlikely an explanation based in merely a projection effect of the four objects.

5. Summary

- New observations of a system of anomalous redshift are presented. These comprises, broad band, and intermediate resolution spectroscopy of four close objects.
- The redshifts of the main galaxy (0.1239 ± 0.0005) and the two previously known compact objects (0.1935 ± 0.0002 and 0.1939 ± 0.0002) have been confirmed with better accuracy.
- We have delineated the diffuse emission apparently connecting the main galaxy and the three compact objects up to the isophote $26.8 \text{ mag arcsec}^{-2}$ in the SDSS r' filter.
- The redshift of the third compact object has been estimated as 0.2229 ± 0.0002 .

- We have shown that the filament probably has a redshift of $z = 0.19$ and seems to be mainly associated with the compact objects.
- From the ratio of several emission lines, we have classified the three emission line objects as a QSO (object 1) and HII galaxies (objects 2 and 3)
- The probability of the whole aggregation being fortuitous has been computed as $\sim 10^{-6}$.

We want to thank Fernanda Artíguez Carro, who kindly made the observations at the IAC 80 that allow a relative calibration of the image taken in the r' filter with the NOT telescope.

REFERENCES

- Arp, H. 1977, ApJ, 218, 70
- Bahcall, J. N., Kirhakos, S., Saxe, D. H., & Schneider, D. P. 1997, ApJ, 479, 642
- Bertin, E., & Arnouts, S. 1996, A&A, 117, 393
- Burbidge, G. 1996, A&A, 309, 9
- Carter, B. J., Fabricant, D. G., Geller, M. J., Kurtz, M. J., & McLean, B. 2001, ApJ, 559, 606
- Canalizo, G., & Stockton, A. 2001, ApJ, 555, 719
- Ellis, R., Santos, M. R., Kneib, J. P., & Kuijken, K. 2001, ApJ, 560, L119
- López-Corredoira, M., & Gutiérrez, C. M. 2003, A&A (submitted)
- López-Corredoira, M., & Gutiérrez, C. M. 2002, A&A, 390, L15
- Metcalfe, N., Shanks, T., Fong, R., & Jones, L. R. 1991, MNRAS, 249, 498
- Smith, J. A. et al. 2002, AJ, 123, 2121
- Stockton, A. 1982, ApJ, 257, 335
- Sulentic, J. W. & Arp, H. 1976, *first year report on NSF grant*
- Verón, P., Goncalves, A. C., & Verón-Cetty, M.-P. 1997, A&A, 319, 52

TABLES

Table 1. Observations

Imaging		
Filter	Exp. time (s)	Seeing (arcsec)
u'	2×1800	2.7
g'	1×900	2.7
r'	5×1800	1.4
Spectroscopy		
Position	Exp. time (s)	Slit width
1	4×1800	3.0
2	1×1800	1.2
3	1×1800	1.2

Table 2. Emission lines

Line	EW (Å)		
	Object 1	Object 2	Object 3
OII ($\lambda 3727$)	26	28	37
NeIII ($\lambda 3869$)	17	11	
H γ	6	3	
H β	7	14	8
OIII ($\lambda 4959$)	41	2	2
OIII ($\lambda 5007$)	117	8	5
NI($\lambda 5198, 5200$)		2	
HeI($\lambda 5876$)	1	2	
OI ($\lambda 6300$)	6	4	
NII ($\lambda 6548$)	2	8	5
H α	10	66	52
NII ($\lambda 6584$)	4	28	17
SII ($\lambda 6717$)	6	16	10
SII ($\lambda 6731$)	7	13	3

Table 3. Density of objects

Object	r (mag)	N (degree ⁻²)
2	19.6	1500
3	20.2	2500
4	17.3	200

FIGURE CAPTIONS

1. A false-colour scale in the r' band of the region around the system NEQ3 (only a $\sim 40'' \times 40''$ section has been plotted). The plot also show the position of the slits (see main text for the details). The objects studied in this article are labelled as 1–4.
2. A grey-scale image in the r' band of the three compact objects, the main galaxy and the filament apparently connecting them. The contours correspond to 25.6, 26.0 and 26.8 mag arcsec $^{-2}$. North is to the right and east to the top.
3. Main spectral features (corrected of redshift) of the three emission line objects analysed in this paper.





